Group ID	12
Student ID	230042**119**, 230042**140**, 230042**146**
Title	**Khoj**
Brief Description	A room-finding web application that connects IUT students with verified homeowners near campus (Board Bazar, Gazipur). Students can register, search and filter rooms by rent, distance from IUT, amenities and room type, view listings on an interactive map, wishlist rooms, chat with owners in-app, and rate and review rooms after their stay. Homeowners register by submitting ownership documents that are verified by moderators, then post and manage listings with photos, pricing, and availability.

Serial	FR ID	Functional Requirement	User Story ID	US Title	As a (precise role)	I want (requirement)	So that (completion condition)	User Story	Acceptance Criteria	Priority	Status
1	FR_1	The system shall allow students to register and log in securely using their IUT email.	US_1	IUT Student Registration	Student	to register using my IUT email with varification	only genuine IUT student can get access	As a Student, I want to register using my IUT email with OTP verification, so that only genuine IUT students can access the platform.	- IUT email domain (@iut-dhaka.edu) is validated via Google Login. - An OTP is sent to the IUT email. - Account is activated only after correct OTP entry. - Invalid or expired OTP shows an error message.	High	In Progress
2	FR_1	The system shall allow students to register and log in securely using their IUT email.	US_2	Secure Login Session	Student	to login and stay securely signed in	My account and personal data remain protected	As a Student, I want to log in and stay securely signed in with a session token, so that my account and personal data remain protected.	- Login requires correct email and password. - A JWT session token is issued on successful login. - Wrong credentials show an error without revealing which field is wrong. - Session expires after the configured time.	High	Done
3	FR_2	The system shall homeowner to register by submitting a home ownership document for Moderator verification	US_3	Home Owner Registration via Ownership Documentation	Home Owner	to register by uploading my home owner certificate	I can be verified and post authentic listings	As a Home Owner, I want to apply for an account by uploading my home ownership document, so that I can be verified and post authentic listings.	- Ownership document upload (image/PDF) is required to submit. - Application status shows as Pending after submission. - Account can post listings only after moderator approval. - Rejected applications show a rejection reason.	High	Done
4	FR_2	The system shall allow admin to validate home owner's registration against the document	US_4	Verify Home Owner documents	Admin	to review submitted documents to approve or reject Registration	Only genuine home owner can get access and post for their listings	As an Admin, I want to review submitted ownership documents and approve or reject applications, so that only genuine homeowners can list rooms on the platform.	- Pending applications appear in the moderation queue. - Uploaded document is viewable before decision. - Approve and Reject actions are available with a reason field. - Applicant is notified of the decision.	High	Done
5	FR_2	The system shall allow Students to search and filter through listings	US_5	Filter rooms by Rent Range	Student	to filter room by a minimum and maximum range	I can only seem room within my budget	As a Student, I want to filter listings by a minimum and maximum rent, so that I can only see rooms within my budget.	- Minimum and maximum rent inputs are available. - Results only include listings within the selected range. - Filter can be cleared to show all listings. - Result count updates after applying the filter.	High	To Do
6	FR_3	The system shall allow Students to search and filter through listings	US_6	Filter rooms by distance, ammenities	Student	to filter room by proximity yo IUT , and available services	I can find room matching my specification	As a Student, I want to filter rooms by proximity to IUT, amenities , so that I can quickly find rooms matching my exact needs.	- Distance-from-IUT filter is available. - Amenity tags (WiFi, AC, generator, lift, etc.) can be selected. - Room type (single, shared, sublet, family, mess-seat) can be selected. - Multiple filters can be combined in one search.	High	In Progress
7	FR_4	The System shall show room locations, and distance from IUT on an interactive map	US_7	View rooms on an interactive map	Student	to see the listings as pin on a map, specifying exact location and see the distance from IUT Gate	I can judge how far and convinient each room's location is	As a Student, I want to see listings as pins on a map with their distance from the IUT gate, so that I can judge how convenient each room's location is.	- Each listing appears as a pin with a price bubble on the map. - Clicking a pin opens a listing popup with details. - Distance from IUT entrance gate is shown for each listing. - Map and result list stay in sync in split view.	Medium	In Progress
8	FR_5	The system shall should each room's availability in real time	US_8	See Real time availability	Student	to see a whether a room is vacant, soon to be vacant or rented	I don't need to manually contact for these information	As a Student, I want to see whether a room is available, soon-to-be vacant, or rented, so that I do not waste time contacting owners of occupied rooms.	- Status (Available / Soon Vacant / Rented) is shown on each listing card. - Status updates without needing a page reload. - Rented rooms are visually distinguished from available ones. - Owner status changes are reflected to students immediately.	Medium	In Progress
9	FR_6	The system should allow students to wishlist rooms, and notify when available	US_9	Wishlist a room	Student	to save room to my wishlist	I can track and revisit rooms I like.	As a Student, I want to save rooms to a wishlist, so that I can track rooms I like and revisit them later.	- A wishlist button is available on every listing. - Wishlisted rooms appear on the student's wishlist page. - Rooms can be removed from the wishlist. - Wishlist persists across sessions.	Medium	To Do
10	FR_6	The system should allow students to wishlist rooms, and notify when available	US_10	Notify when wishlisted room is vacant	Student	to receive a notification when a wishlisted room becomes available for rent	I can quickly get notified and act before the room is taken	As a Student, I want to receive a notification when a wishlisted room becomes available for rent, so that I can act quickly before the room is taken.	- Notification is generated when a wishlisted room's status becomes Available. - Notification links to the listing. - Notifications are visible in the app. - No notification is sent for rooms removed from the wishlist.	Medium	To Do
11	FR_7	The system shall allow homeowners to create and manage room listings.	US_11	Post a room listing	Home Owner	to post a room with rent, location, room type, minimum stay and amenity tags	students can find and evaluate my room accurately	As a Home Owner, I want to post a room with rent, location, room type, minimum stay and amenity tags, so that I canstudents can find and evaluate my room accurately	- Listing form requires rent, location (map pin), and room type. - Amenity tags can be attached to the listing. - Minimum rent duration can be specified. - Listing appears in search results after submission/approval.	High	In Progress
12	FR_7	The system shall allow homeowners to create and manage room listings.	US_12	Upload categorized room photos	Home Owner	to upload photos of the room organized by category (room, bathroom, balcony, kitchen)	students can inspect the condition of each part of the flat remotely	As a Home Owner, I want to upload photos of the room organized by category (room, bathroom, balcony, kitchen), so that I canstudents can inspect the condition of each part of the flat remotely	-Multiple photos can be uploaded per listing. - Photos are displayed by category on the listing page. - Unsupported file types are rejected with an error	Medium	Done

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13	FR_7	The system shall allow homeowners to create and manage room listings.	US_13	Manage availability of listings	Home Owner	to toggle a listing between available and rented	students always see the correct status of my rooms	As a Home Owner, I want to toggle a listing between available, soon-vacant and rented, so that students always see the correct status of my rooms.	- Status can be changed from the My Listings page. - Status change is reflected in student search immediately. - Multiple listings can be managed from one dashboard. - Listings can be edited or deleted by their owner only.	High	In Progress
14	FR_8	The system shall provide real-time in-app chat between students and homeowners.	US_14	Chat with homeowner	Student	to chat with a homeowner directly inside the app	I can negotiate rent and clarify rules before visiting	As a Student, I want to chat with a homeowner directly inside the app, so that I can negotiate rent and clarify rules before visiting.	- Chat can be started from any listing page. - Messages are delivered in real time (Socket.IO). - Chat history is preserved across sessions. - Unread messages are indicated to the user.	High	In Progress
15	FR_8	The system shall provide real-time in-app chat between students and homeowners.	US_15	Respond to student inquiries	Home Owner	to receive and reply to student messages about my listings	I can answer questions and finalize rental agreements quickly	As a Home Owner, I want to receive and reply to student messages about my listings, so that I can answer questions and finalize rental agreements quickly.	- Incoming messages show which listing they relate to. - Owner can reply from a single inbox for all listings. - New message notifications are shown to the owner. - Conversations are only visible to the two participants.	High	In Progress
16	FR_9	The system shall allow students to rate and review rooms after their stay, and homeowners to respond.	US_16	Rate and review a room	Student	to rate a room per category (cleanliness, location, value, safety, accuracy, communication) after renting it	future students can make informed decisions from verified feedback	As a Student, I want to rate a room per category (cleanliness, location, value, safety, accuracy, communication) after renting it, so that future students can make informed decisions from verified feedback.	- Review is allowed only for students who rented the room. - Each rating category accepts a 1-5 score. - A written review can accompany the ratings. - The listing shows the average rating and category breakdown.	High	In Progress
17	FR_10	The system shall allow students to report fake or misleading listings, and moderators to act on those reports.	US_17	Report a fake listing	Student	to report a listing that is wrong, fake or misleading with a reason	other students are protected from fraudulent listings	As a Student, I want to report a listing that is wrong, fake or misleading with a reason, so that other students are protected from fraudulent listings.	- A report button with a reason field is available on every listing. - Duplicate reports from the same student are prevented. - Reporter identity is hidden from the homeowner. - Report confirmation is shown after submission.	Low	To Do
18	FR_10	The system shall allow students to report fake or misleading listings, and moderators to act on those reports.	US_18	Handle reported listings	Admin	to review reported listings and remove or restore them	the platform stays free of fake and misleading content	As a Moderator, I want to review reported listings and remove or restore them, so that the platform stays free of fake and misleading content.	- Reported listings appear in a moderation queue with reasons. - Moderator can view the full listing and report details. - Moderator can remove the listing or dismiss the report. - The action taken is recorded with the moderator's ID.	Low	To Do
19	FR_11	The system shall allow students to post roommate advertisements for shared rooms where the homeowner permits it.	US_19	Advertise for a roommate	Student	to post a roommate advertisement for my shared room	I can find a compatible roommate and split the rent	As a Student, I want to post a roommate advertisement for my shared room, so that I can find a compatible roommate and split the rent.	- Roommate ad can be created only if the owner enabled sharing. - Ad includes room reference, rent share and preferences. - Other students can view and respond to the ad. - The ad can be closed once a roommate is found.	Low	In Progress
20	FR_12	The system shall allow admins to manage users, roles and platform configuration.	US_20	Manage users and roles	Admin	to view all users and assign or revoke roles (student, home owner, moderator)	the platform is governed with correct permissions	As a Admin, I want to view all users and assign or revoke roles (student, home owner, moderator), so that the platform is governed with correct permissions.	- Admin dashboard lists all users with their roles. - Roles can be assigned or revoked per user. - Suspicious accounts can be suspended. - All admin actions are logged.	High	To Do